

Beyond Traditional Charts: How Key Influencers & Top Segments Drive Business Decisions.



Project Scope & Core Data

- Dataset Scope: Analyzed comprehensive housing records spanning multiple market regions with a Total Sale Value of \$3.10bn.
- Primary Goal: Uncover hidden cross-functional mathematical patterns that dictate why property prices spike High or drop Low.
- Advanced Engineering: Formulated dedicated DAX layers (**Built Category** calculated column & **Year Over Year Growth %** measure) to capture structural market behaviors and macro-level performance trends.

```
1 Built Category =  
2 VAR Time_diff = YEAR(housing_data[date]) - housing_data[year_build]  
3  
4 RETURN  
5     SWITCH(TRUE(),  
6         ISBLANK(housing_data[year_build]), "Unknown",  
7         Time_diff < 0, "Pre-Built (purchased before built)",  
8         Time_diff <= 5, "Brand New (0 to 5 years)",  
9         Time_diff <= 15, "Recent Built (6 to 15 years)",  
10        "Established (+16 years)"  
11    )
```

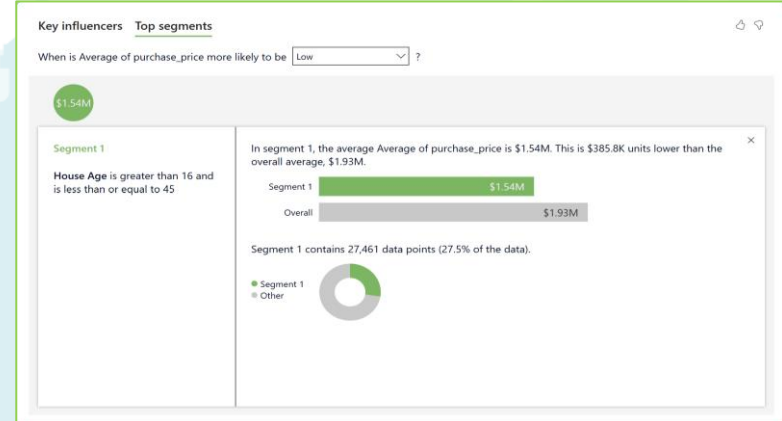
```
1 Year Over Year Growth % =  
2 VAR Cur_Yr_Rev = SUM(housing_data[purchase_price])  
3 VAR Pvs_Yr_Rev = CALCULATE(SUM(housing_data[purchase_price]), SAMEPERIODLASTYEAR(DATESYTD(Date_Table[Date])))  
4  
5 RETURN  
6 IF(Pvs_Yr_Rev = 0 || Not(ISBLANK(Pvs_Yr_Rev)), DIVIDE(Cur_Yr_Rev - Pvs_Yr_Rev, Pvs_Yr_Rev), Blank())
```

The Power of Key Influencers Visual

- Advanced Machine Learning: Automatically executes complex regression diagnostics and linear/logistic validation behind the scenes without manual coding.
- Granular Analysis Optimization: Structured visual analytics using precise single-row grain filtering via `house_id` to eliminate data conflation errors.
- Actionable Segmentation: Segregates massive noise instantly into distinct, statistically relevant 'Top Segments' that corporate stakeholders can query dynamically.

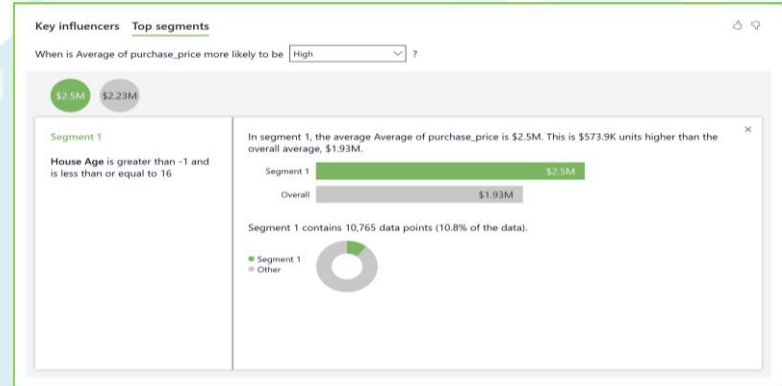
Insight 1 – What Drives Prices 'LOW'?

- Core AI Finding: Real estate values experience structural mid-life depreciation heavily driven by property maturity timelines.
- Statistical Boundary: House Age is greater than 16 and less than or equal to 45 years.
- High-Impact Segment: Segment 1 holds 27,461 specific data points (27.5% of dataset) with an average price of \$1.54M.
- Baseline Variation: Drops overall property value significantly by \$385.8K lower than the baseline market average (\$1.93M).



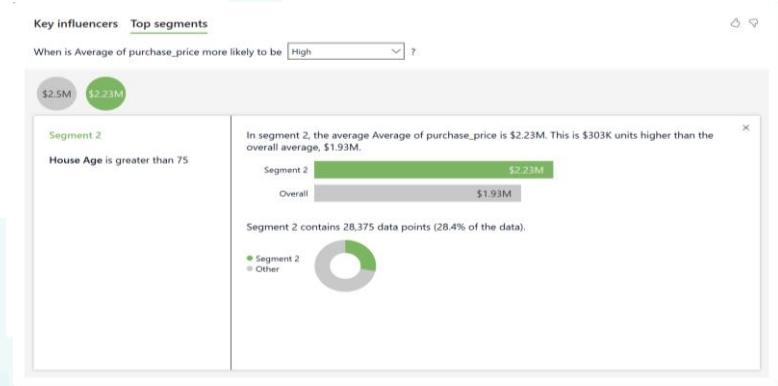
Insight 2 – What Drives Prices 'HIGH'?

- Core AI Finding: Maximum real estate premium metrics thrive dynamically on brand-new infrastructure windows.
- Statistical Boundary: House Age is less than or equal to 16 years (Brand New / Newly Built).
- High-Impact Segment: Segment 1 surges value to an average price of \$2.50M, capturing 10,765 targeted market observations.
- Premium Metrics: Forces property values to climb an impressive \$573.9K higher than the general baseline average (\$1.93M).



Insight 3 – The Vintage Premium

- Core AI Finding: Property values do not depreciate indefinitely; they experience a distinctive 'U-Curve' validation curve as infrastructure shifts to historical status.
- Statistical Boundary: House Age is greater than 75 years (Vintage / Historical Properties).
- High-Impact Segment: Segment 2 encompasses 28,375 rich data points with a stabilized premium average price of \$2.23M.
- Asset Valuation: Drives property value evaluations to sit exactly \$303K higher than the cumulative market baseline average.



Enterprise-Grade Data Storytelling

- Strategic Cross-Filtering: Coupled native Power BI slicers directly to dynamic dimension tables to ensure immediate live regional data updates.
- Automated AI Narration: Integrated native Smart Narrative tokens to compile natural language analytics directly alongside complex data visuals.
- Direct Financial Utility: Empowers real estate investment groups to target the 16–45 year window for massive discount buys or capture high premium yields within new builds.

Let's Connect & Innovate!

- Open-Source Architecture: The full interactive .pbix model framework, schema data steps, and documentation are deployed on my GitHub profile.
- Professional Network Engagement: How are you implementing automated machine learning visuals into your custom workspace pipelines? Let's discuss in the comments below!
- Contact Framework: Paing Soe Tun | LinkedIn [[Link](#)]